Coding workshop 1

CS 4824/ECE 442

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Rules

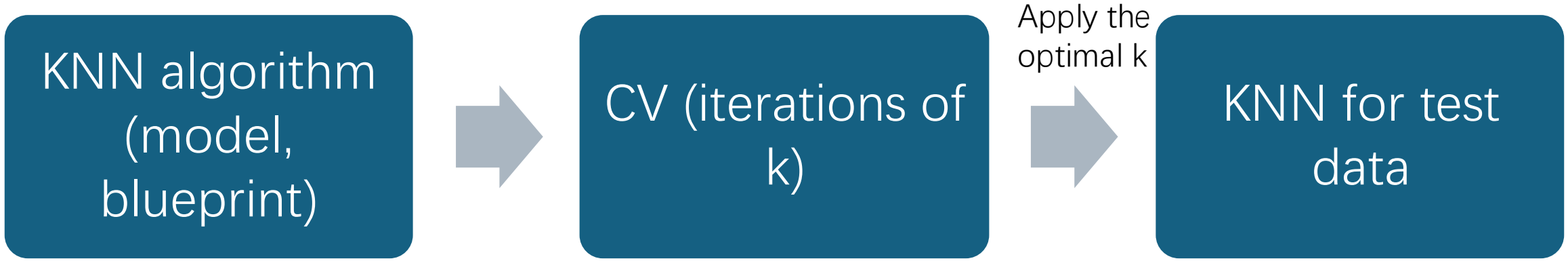
- **NO:** sklearn.neighbors, sklearn.linear_model, sklearn.model_selection
- **YES:** numpy (Matrix computation), pandas (Read CSV), collections (Counter, votes), matplotlib (Plots)
- Understand the mathematical foundations

KNN

- **Task:** determine 5 or 6
- **Dataset:** vectors, row (picture), column (attribute, Black pixel count per 4×4 patch)
- **Core:** Euclidean distance & Votes (k, most common)
- **Tie-breaking:** "*Break ties at random*" : np.random.choice

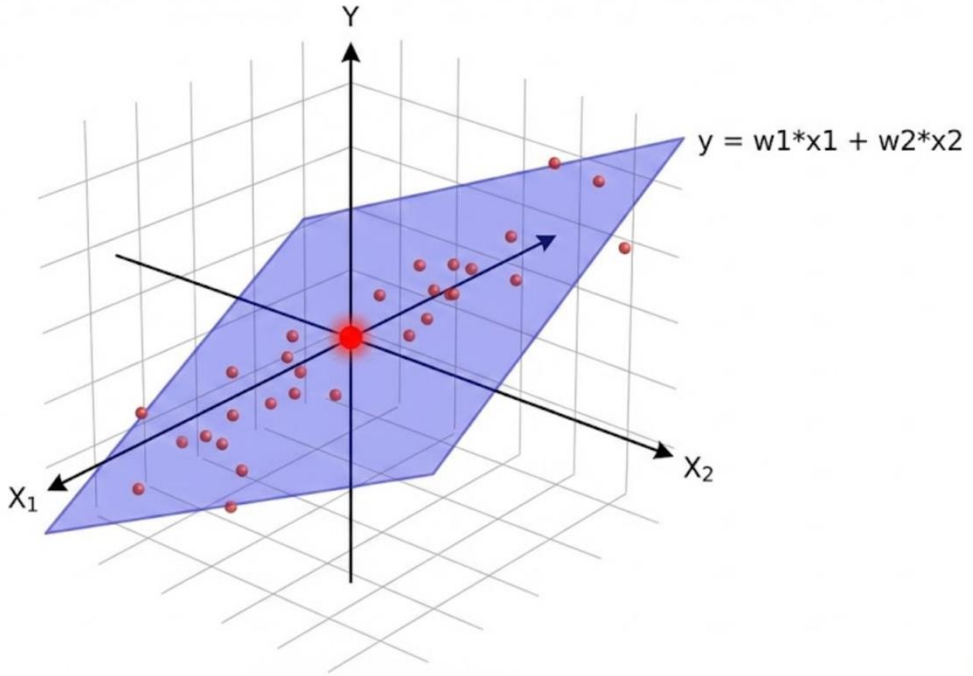
KNN

- To determine the best k, apply it in KNN
- No data leakage
- Cross Validation: read in every loop? pd.concat (a complete train dataset)
- !!! reset_index(drop=True)
- Accuracy vector and mean value



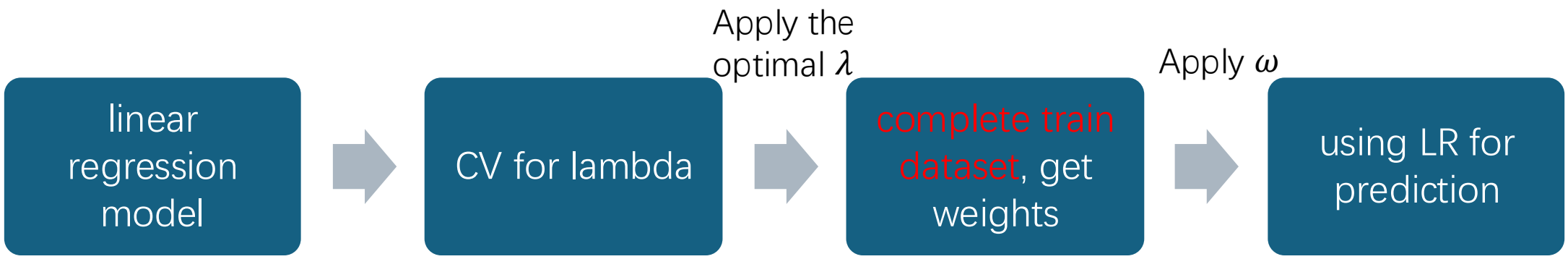
Linear Regression

- $L(\omega) = \frac{1}{m}\sum_{n=1}^{m}(x_n\omega - y_n)^2 + 0.5\lambda\omega^T\omega$
- ★ $L(\omega) = \frac{1}{m}\sum_{n=1}^{m} r_n(x_n\omega + b - y_n)^2$
- **Task:** Approximation
- **Core:** Gradient descent(matrix multiplication, weights renewal, learning rate α (0.001?))



Linear Regression

- Cross Validation: to determine the best λ
- λ affects the performances, the weights ω
- Get the corresponding ω
- Use ω on test dataset



Comparisons

	Hyperparameter	Parameter
KNN	k	(No parameter, only counting & voting)
LR	λ(And α)	ω(if you have bias b)

Brainstorm

```
sum_x = np.sum(X**2, axis=1, keepdims=True)
sum_y = np.sum(Y**2, axis=1).reshape(1, -1)
inner_product = np.dot(X, Y.T)
dists = np.sqrt(sum_x + sum_y - 2 * inner_product)
```

- In KNN, if test dataset size is M, train dataset size is N, then a double loop calculating distances will be $M \times N$ rounds. We all know loop is always time-consuming. How can you simplify it? (**Hint:** matrix, numpy broadcasting)
- What does "accuracy" in the linear regression instructions mean? (**Hint:** metric selection)

$$\omega = (X^T X + \lambda I)^{-1} X^T Y$$

- Why does adding $0.5\lambda\omega^T\omega$ make the optimization problem "well-posed" even when $X^T X$ is singular? (**Hint:** closed-form solution)
- What will happen if we don't ignore b when coding linear regression? (**Hint:** try to do something about X)

$$Y = (X, 1)(\omega, b)^T$$

Thank you!